

FIELD THEORY PART-II

Phase 6: Magnetic Boundary Conditions

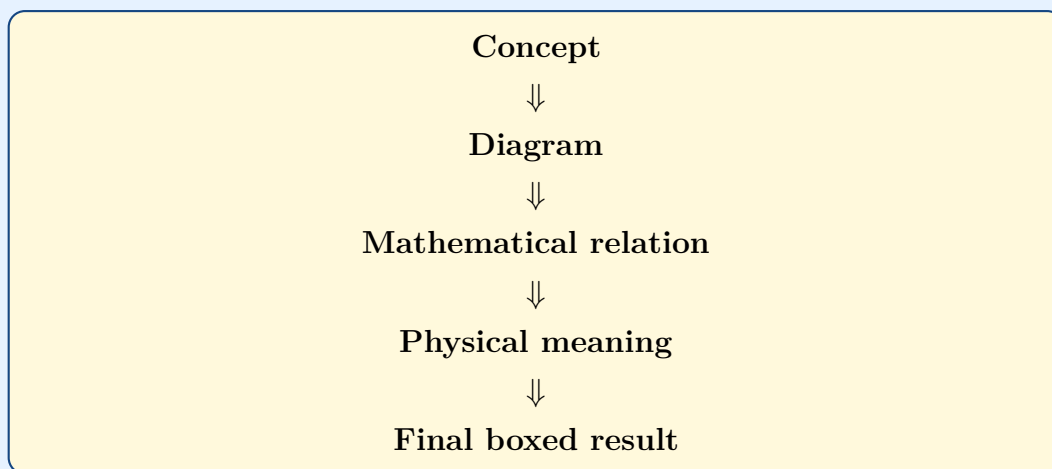
Tangential H, normal B and magnetic refraction relation

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 6

Main Goal

This phase prepares the repeated long-answer topic: magnetic boundary conditions between two magnetic media.

2 Problem Setup

Two Magnetic Media

Consider two magnetic media separated by a plane boundary. Their permeabilities are:

$$\mu_1, \quad \mu_2.$$

Magnetic flux density and magnetic field intensity are related by:

$$\boxed{\mathbf{B}_1 = \mu_1 \mathbf{H}_1, \quad \mathbf{B}_2 = \mu_2 \mathbf{H}_2}$$

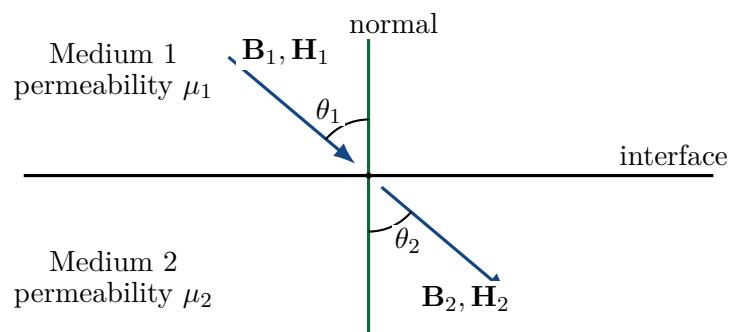


Figure 1: Magnetic field passing through two magnetic media.

3 Tangential Component of H

Using Ampere Law

Apply Ampere's circuital law to a small rectangular loop crossing the boundary:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

If no surface current exists:

$$I_{\text{enc}} = 0.$$

So:

$$H_{1t}\Delta l - H_{2t}\Delta l = 0.$$

Therefore:

$$\boxed{H_{1t} = H_{2t}}$$

If angles are measured from normal:

$$H_1 \sin \theta_1 = H_2 \sin \theta_2$$

4 Normal Component of B

Using Gauss Law for Magnetism

Gauss law for magnetism is:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0.$$

Apply it to a small pillbox across the boundary:

$$B_{1n} \Delta S - B_{2n} \Delta S = 0.$$

Therefore:

$$B_{1n} = B_{2n}$$

If angles are measured from normal:

$$B_1 \cos \theta_1 = B_2 \cos \theta_2$$

5 Magnetic Refraction Relation

Combining the Conditions

Using:

$$B_1 = \mu_1 H_1, \quad B_2 = \mu_2 H_2,$$

the normal condition becomes:

$$\mu_1 H_1 \cos \theta_1 = \mu_2 H_2 \cos \theta_2.$$

The tangential condition is:

$$H_1 \sin \theta_1 = H_2 \sin \theta_2.$$

Dividing properly:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

6 Application

Transformer Core

In transformer cores, iron has very high permeability. Magnetic flux prefers to pass through high-permeability material because magnetic reluctance is low. Thus magnetic boundary condition explains why flux is guided through iron cores.

7 Exam Writing Template

Answer Pattern

Write:

1. Draw interface diagram.
2. State $B = \mu H$ in both media.
3. Use Ampere law for tangential H .
4. Use Gauss law for normal B .
5. Combine to get $\tan \theta_1 / \tan \theta_2 = \mu_1 / \mu_2$.
6. Give application.

8 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.